MACHINE LEARNING UNIVERSITY

EEP-AGESS - Agent Essentials Design Doc

Basic information

Course code name: MLU-EEP-AGAI-M1

Course name: **Agentic AI: Essential Concepts for Builders**

Projected full duration: 4 hours (Can vary depending on modularization based on students' personas)

Short course description

Agent Essentials is a code-free lesson that introduces faculties to the emerging field of Agentic AI and explores how AI agents use LLMs to think and plan, then execute actions through connected tools—creating a cycle of reasoning, deciding, acting, and learning from results. The lesson details theoretical foundations of agentic systems along with standard terminology used in the field. The lesson compares traditional generative AI methods with agentic solutions, enabling faculties to differentiate between both approaches and to understand when to use agents, how to design agent-based solutions, and how to evaluate AI agents.

Long course description:

As AI continues to evolve, there's a growing need to understand the power of AI systems that can reason, act, and make decisions autonomously. Traditional approaches to AI implementation often fall short when dealing with complex, multi-step tasks that require reasoning, planning, and adaptation. This lesson was developed to address the fundamental gap between simple generative AI applications and more sophisticated agentic systems that can understand context, maintain goal-oriented behavior, and operate with greater autonomy. Whether the faculty teaches code-based or code-free courses, understanding agentic AI has become crucial in today's rapidly evolving professional landscape. The lesson provides essential knowledge for faculties who need to distinguish between different AI approaches and prepare themselves to make informed decisions about when and how to implement agentic solutions.

They will engage in activities and real-world examples to master the fundamentals of agentic AI systems without writing code. Faculties will analyze case studies of successful agentic AI implementations, practice agent design thinking through interactive discussions, and understand evaluation metrics for assessing agent performance. The lesson includes collaborative discussions where faculties will learn about various real-world use cases and how to adapt them to different professional contexts. By the lesson's conclusion, faculties will be able to explain the fundamental concepts of Agentic AI, including their key principles, architectures, and design considerations. They will be able to analyze various business and technical use cases to determine whether an agent-based implementation would be suitable and beneficial for a project. Faculty will be able to differentiate between scenarios that require agent-based approaches versus those better served by traditional non-agent solutions.

How challenging is this course?

Difficulty level: 100

What category does this course fall into?

Agentic AI

Profiling information

Learners

Target Audience: The audience for the EEP program is faculty in higher education institutions as part of MLU Education Enablement Program, who want to integrate Agentic AI content into their skills and knowledge, preparing for teaching this content.

To make the current **MLU AGESS – Agent Essentials** content accessible to community college faculties, we must account for their diverse backgrounds—across both code-based and code-free programs. By creating tailored user stories for these two distinct faculty personas, the content can be adapted to address their unique learning needs and teaching goals.

These persona-based user stories will also support content modularization, giving faculty members the flexibility to choose modules that are most relevant to their students' personas and educational backgrounds.

The following user stories, based on these two personas and this learning material, define what each learner should be able to accomplish upon course completion:

code-based faculty

This faculty prepares and teaches courses for degrees in hard sciences, engineering, mathematics, or related technical disciplines such as physics, computer science, or mechanical engineering. Their software development experience ranges from beginner to intermediate—some may have completed programming courses or personal projects, while others are just starting. They're comfortable with fundamental coding concepts like loops, functions, and conditionals, as well as data structures including arrays, lists, and trees.

code-free faculty

This faculty prepares and teaches courses for degrees in business, healthcare administration, liberal arts, or similar non-technical disciplines. They possess strong analytical and critical thinking abilities and recognize technology's influence on their field, yet their technical knowledge remains limited to everyday consumer applications. Relevant examples include soft science and business courses such as social sciences or health sciences. While technical subjects may seem daunting, they're eager to grasp AI technology fundamentals and practical applications.

Content Modularization

Lecture activities

All MLU EEP course content is modularized, allowing faculty members the flexibility to choose modules that are most relevant to their educational backgrounds.
Throughout our course's slide deck, we use two different icons to identify the proposed target persona relate to the content:

code-based Course Icon

code-free Course Icon

Both lecture content, supported by slide decks, and lab hands-on material are identified by these icons.

Usage example

If you your background is code-free, look for all slides that have this icon:

otherwise, look for all slides that have this icon:

Hands-on activities

Hands-on activities also are modularized considering three different personas, adding two categories to the code-based personas:

code-free faculty

These hands-on activities aim to give experience creating AI solutions using Amazon Quick. This no-code environment is ideal for faculties without programming background, enabling them to build sophisticated AI solutions effortlessly.

User Stories

id	User Stories, what do you want the people to be able to do when they leave the class? As a ___, I want ___ so that ___
Code-based faculty 	As a **code-based faculty**, I want to understand the theoretical foundations and principles of agentic AI systems so that I can apply this knowledge in advanced programming projects and prepare for a career in AI development.
	As a **code-based faculty**, I want to understand the key architectural components of agentic AI systems so that I can build my own AI-powered applications and contribute to open-source projects.
	As a **code-based faculty** I want to understand how to integrate agentic AI with existing software systems so that I can develop practical solutions for real-world problems in my capstone projects.
	As a **code-based faculty**, I want to understand the capabilities and limitations of agentic AI so that I can make informed technical decisions when designing my own AI applications.
	As a **code-based faculty**, I want to understand what agentic AI is and how it differs from regular AI so that I can decide if this is a field I want to pursue in my studies.
	As a **code-based faculty**, I want to see practical examples of how agentic AI is used in everyday applications so that I can understand its relevance to my future career.

	As a **code-based faculty**, I want to understand when to use agentic AI versus simpler solutions so that I can develop good problem-solving habits early in my education.
Code-free faculty	As a **code-free faculty**, I want to understand the business value and strategic implications of agentic AI so that I can recommend AI solutions in my future management or consulting role.
	As a **code-free faculty**, I want to understand how agentic AI systems work at a conceptual level so that I can effectively communicate with technical teams and stakeholders in my future career.
	As a **code-free faculty**, I want to analyze case studies of successful agentic AI implementations so that I can identify opportunities for AI adoption in various business contexts.
	As a **code-free faculty**, I want to understand the capabilities and limitations of agentic AI so that I can set realistic expectations and make informed recommendations in my professional projects.
	As a **code-free faculty**, I want to understand what agentic AI is in simple terms so that I can participate in conversations about emerging technologies in my field.
	As a **code-free faculty**, I want to see real-world examples of agentic AI applications so that I can understand how this technology might impact my future career.
All personas	As a **faculty**, I want to learn about the ethical considerations and limitations of agentic AI so that I can be a responsible user and advocate for appropriate AI use in my profession.

Learner Outcomes

id	Action verb (After completing this course, learners are prepared to...)	(finish the sentence)	How will faculties demonstrate they have learned this in the course? (Emphasis on demonstrate)
1	Explain	fundamental concepts of Agentic AI, including their key principles, architectures, and design considerations for teaching purposes.	Lecture, Discussion, Quiz
2	Analyze	various business and technical use cases to determine whether an agent-based implementation would be suitable and beneficial for projects.	Lecture, Case Study Analysis, Quiz
3	Differentiate	between scenarios that require agent-based approaches versus those better served by traditional non-agent solutions.	Lecture, Discussion Activities, Quiz
4	Explain	the role of Responsible AI and Agentic Evaluation in the Agentic world and how to teach these concepts.	Lecture, Discussion, Quiz
5	Explain	limitations and capabilities of various categories of agentic landscapes and orchestration patterns.	Lecture, Comparative Analysis, Quiz

6	Explain	The scope, purpose, and impact of various agentic implementations in real-world contexts and how to adapt them to educational settings.	Discussion, Case Studies, Quiz
7	Apply	Practice with Amazon Quick to build an AI-powered tool that support academic success and study activities	Demonstration, Guided activities

Prerequisites

Understanding of Gen AI fundamentals	**Required**	Need to understand how LLMs process and generate text; Knowledge of token limits and their implications; Understanding of model capabilities and limitations; Ability to choose appropriate models for specific tasks; Understanding temperature and other parameter settings
Understanding of Basic Prompting techniques like ReAct	**Optional**	**Importance for Agents:** Enables agents to access and utilize external knowledge; Crucial for grounding agent responses in factual information; Helps mitigate hallucinations and improve accuracy; Provides a framework for structured reasoning and action planning; Enables step-by-step problem solving and decision making; Crucial for creating agents that can break down complex tasks.

Completion Criteria

Completion criteria will be defined by faculty members using this content. Below are some suggestions.

Quiz/Evaluation	Questions covering the learning outcomes like agent basics, when and why to use agents, evaluation frameworks, and responsible AI considerations.
Participation	Active engagement in discussions about use cases, orchestration patterns, and real-world applications
Case Analysis	Successful adaptation of technical concepts for different faculty personas and skill levels by building an AI agentic app.

Detailed topic list

Flexibility is a key driver for implementing MLU EEP content. Faculty members will adapt the content according to their own instructional plans. The table below provides general suggestions for topic coverage aligned with the learning objectives.

Activity #	Name	Topic covered	Additional details	Learner outcome
Section 1	AGI and Agentic AI	Introduction to Agentic AI.	Distinguish between AGI concepts and practical	**LO1:** Explain fundamental concepts of Agentic AI

		What are LLM agents?	agentic AI systems; Core principles of agents	
Section 2	Should I use or not use agents?	Do I need agents? What problems do they solve? What are my options?	Discuss current level of understanding and use of Agents in teaching contexts	**LO2:** Analyze use cases to determine agent implementation suitability **LO3**: Differentiate between agent-based and non-agent approaches
Section 3	Responsible AI and Agentic Evaluation	Agentic Evaluation and RAI overview	Different levels of metrics for Agentic Evaluation and Responsible AI considerations when working with agents; Teaching RAI in agentic contexts	**LO4**: Explain role of Responsible AI and Agentic Evaluation in the Agentic world
Section 4	Agentic Landscape and Orchestration Patterns	Current Landscape of Agentic Platforms and orchestration patterns	High level overview and appropriateness, pros and cons for low/no-code, code first, domain specific frameworks; Selecting appropriate tools for faculty skill levels.	**LO5**: Explain limitations and capabilities of various categories of agentic landscapes and orchestration patterns

		Agentic Computer Use and Human-Computer Interaction Revolution	High level overview of how agentic computer use, are rapidly reshaping human-computer interaction landscapes	**LO5**: Explain limitations and capabilities of various categories of agentic landscapes and orchestration patterns
Session 5	Agentic AI use-cases and applications	Discuss real-world use-cases and applications	Scope, purpose and impact of various agentic implementations; Adapting use cases for educational contexts and faculty projects	**LO6**: Explain scope, purpose and impact of various agentic implementations
		Understanding no-coding agentic tools	This module covers key concepts and functionalities of a no-code platform to build agentic AI tools. We will present the key functionalities of Amazon Quick Suite, used in the hands-on lab.	**LO7**: Practice with Amazon Quick Suite to build an AI-powered tool that support academic success and study activities
		Lab 1.1	Practice creating custom Chat Agents, and Flows in	**LO7**: Practice with Amazon Quick Suite to

		Creating an AI assistant to support my studying activities	Quick Suite to build AI-powered tool that support study activities accessing relevant data	build an AI-powered tool that support academic success and study activities

Hands-on Activities module

Quick Suite Hands-on lab

Objective: Introduce faculties to Amazon Quick Suite fundamentals, exploring how to build AI-powered tools that support academic study activities.

Lab 1.1: Using Quick Suite chat to support your study activities

Internal reference (remove for final version)

- MLU AGESS course design documentation

Alignment with EEP Goals

This lesson aligns with the MLU Educator Enablement Program's strategic goals:

- **Faculty enablement:** Designed specifically for educators who will teach agentic AI concepts
- **Content adoption:** Provides faculty with foundational knowledge to confidently integrate agentic AI into their curriculum
- **Modular approach:** Part of a multi-lesson sequence that can be taught independently or as a complete module
- **Scalable education:** Prepares faculty to teach hundreds of students, multiplying the impact of the training

- **Industry-relevant skills:** Ensures students learn current, in-demand AI capabilities that prepare them for the workforce

References by session

Session 1 Resources: AGI and Agentic AI

code-based faculty

ReAct: Synergizing Reasoning and Acting in Language Models

Cognitive Architectures for Language Agents

code-free faculty

Agentic AI, explained

Agents for growth: Turning AI promise into impact

Session 2 Resources: Should I use or not use agents?

code-based faculty

On the Brittle Foundations of ReAct Prompting for Agentic Large Language Models

Enhancing LLM Reasoning with Multi-Path Collaborative Reactive and Reflection agents

code-free faculty

To Thrive in the AI Era, Companies Need Agent Managers

Generative AI and Business Transformation

MBAi degree for Business & Artificial Intelligence

Session 3 Resources: Responsible AI and agentic evaluation

code-based faculty

The Measurement Imbalance in Agentic AI Evaluation Undermines Industry Productivity Claims

Adaptive Monitoring and Real-World Evaluation of Agentic AI Systems

Artificial Intelligence (AI) Trust Framework and Maturity Model: Applying an Entropy Lens to Improve Security, Privacy, and Ethical AI

code-free faculty

A Blueprint for Enterprise-Wide Agentic AI Transformation

Session 4 Resources: Agentic Landscape and Orchestration Patterns

code-based faculty

Introducing the Model Context Protocol

Advancing Multi-Agent Systems Through Model Context Protocol: Architecture, Implementation, and Applications

Beyond Accuracy: A Multi-Dimensional Framework for Evaluating Enterprise Agentic AI Systems

the AGNTCY Platform

Alpha Berkeley: A Scalable Framework for the Orchestration of Agentic Systems

Earl: Efficient Agentic Reinforcement Learning Systems for Large Language Models

Efficient and Scalable Agentic AI with Heterogeneous Systems

Tacit Learning with Adaptive Information Selection for Cooperative Multi-Agent Reinforcement Learning

LLM Collaboration With Multi-Agent Reinforcement Learning

The Orchestration of Multi-Agent Systems: Architectures, Protocols, and Enterprise Adoption

A Layered Protocol Architecture for the Internet of Agents

Reasoning-Aware Prompt Orchestration: A Foundation Model for Multi-Agent Language Model Coordination

Agent S: An Open Agentic Framework that Uses Computers Like a Human

Agentic Persona Control and Task State Tracking for Realistic User Simulation in Interactive Scenarios

code-free faculty

One year of agentic AI: Six lessons from the people doing the work

Human-Artificial Interaction in the Age of Agentic AI: A System-Theoretical Approach